



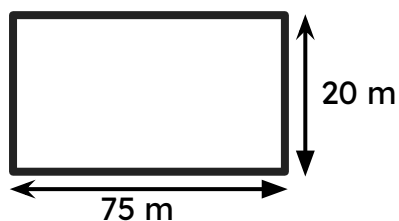
Agricultural worker (animals)

Task A: Geometry in agricultural work

1) For each of the following calculate the maximum number of each animal which can be safely and comfortably looked after in these barns:

Diagrams are not drawn to scale

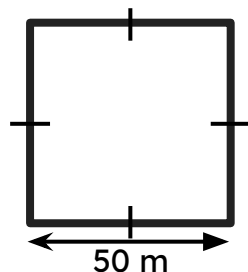
a)



- i) 150
- ii) 1500
- iii) 681
- iv) 750

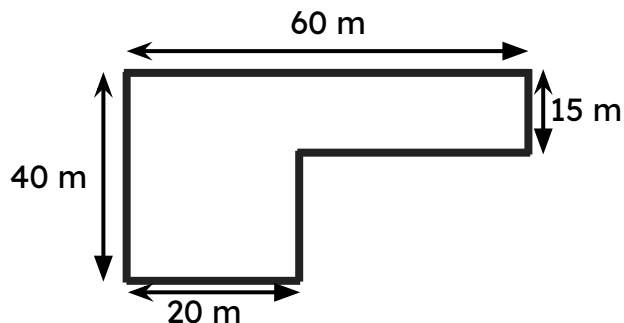
	Animal	Required living area
i	Cows	10 m ²
ii	Chickens	1 m ²
iii	Pigs	2.2 m ²
iv	Sheep	2 m ²

b)



- i) 250
- ii) 2500
- iii) 1136
- iv) 1250

c)



- i) 140
- ii) 1400
- iii) 636
- iv) 700



Task B: Costs and percentages

1) A farmer decides that they want to give their cows more space.

Cost of care for a cow per day is £4

They are going to give each cow 12 m².

Assume each cow produces 20 litres of milk a day.

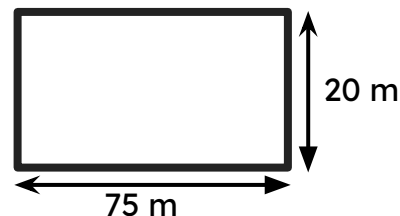
The selling price per litre of milk is 38 p.

Calculate the following:

a) The number of cows the barn can house.

$$\text{Area of barn} = 1500 \text{ m}^2$$

$$\text{Number of cows} = 1500 \div 12 = 125$$



b) The cost of care for the cows per year.

$$\text{Cost of care} = 125 \times 4 \times 365 = \text{£}182\,500$$

c) The income from milk production.

$$\text{Litres of milk} = 125 \times 20 \times 365 = 912\,500$$

$$\text{Income} = 912\,500 \times 0.38 = \text{£}346\,750$$

d) The percentage profit.

	%	£	
	100	182 500	
$\times 1.9$	190	346 750	$\times 1.9$

90% profit is made

2) A farmer decides that they want keep chickens.

They are going to give each chicken an area of 15 m^2

Cost of care for a chicken per year is £50

Each chicken produces 0.8 eggs on average per day.

They decide to sell the eggs themselves, selling 1 dozen for £3.20

Calculate the following:

a) The percentage profit.

$$\text{Area of barn} = 1500 \text{ m}^2$$

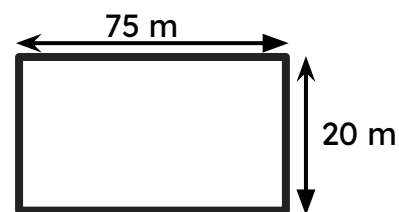
$$\text{Number of chickens} = 1500 \div 15 = 100$$

$$\text{Cost of care per year} = 100 \times 50 = £5000$$

$$\text{Number of eggs per year} = 100 \times 0.8 \times 365 = 29\,200$$

$$\text{Income from eggs per year} = 29\,200 \div 12 \times £3.20 = £7786.67$$

$$\text{Percentage profit} \approx 55.7 \%$$



b) The farmer wants make a 60% profit. How much do they need to charge for a dozen eggs?

$$\text{Cost of care per year} = 100 \times 50 = £5000$$

$$\text{Number of dozens per year} = 29\,200 \div 12 = 2433$$

$$\text{Income for a 60% profit} = £5000 \times 1.6 = £8000$$

$$\text{Cost per dozen} = £8000 \div 2433 = 3.2881... \approx £3.29$$

